Home Lab 5: The Cabbage-Indicator pH Ladder

Do this with a parent. Goggles on. Never taste anything. Clean up after.

Goal

Learn to read pH from the cabbage-indicator color, and watch neutralization happen.

You need

From the kit: **cabbage indicator, baking soda, citric acid** (and cream of tartar if you want an extra). From home: white vinegar, lemon juice, water, milk of magnesia or a little dish soap (optional, for the basic end), 6 clear cups, droppers or spoons.

Part 1: The ladder

Put a little of each liquid in its own cup. Add ~5 drops of cabbage indicator to each. Record the color and estimate the pH from the chart.

Cup	Color	Estimated pH	Acid / Neutral / Base
Vinegar			
Lemon juice			
Water			
Baking soda dissolved in water			
Citric acid dissolved in water			
Dish soap in water (optional)			

Color chart: red ≈ 2 · pink ≈ 4 · purple ≈ 7 · blue ≈ 8 · green ≈ 10 · yellow ≈ 12

Part 2: The color change (neutralization)

1. Fill a cup with water and lots of cabbage indicator (deep purple).
2. Add vinegar a few drops at a time, stirring, until it turns **pink/red**.
3. Now add baking-soda water a little at a time. Watch it climb back: pink → purple → blue → maybe green.
4. Write down every color it passed through.

Follow-up

1. Which cups were acids? Which were bases? Which was neutral?
2. Citric acid is a dry powder. What color did its water solution turn, and what does that tell you about it?

3. In Part 2, what were you doing chemically when the color climbed back up from pink to blue?
4. If a mystery liquid at a competition turned cabbage indicator bright green, what would you know about it?

Coach notes

- Vinegar & lemon juice → red/pink (acid, pH 2–3). Water → purple (neutral). Baking soda water → blue/blue-green (base, pH 8–9). Citric acid water → red/pink (acid, pH ~2–3). Soap → green/blue (base).
- Part 2: adding baking soda (a base) **neutralizes** the vinegar (acid) and then makes the solution basic — acid + base → salt + water.
- Follow-up (4): it's a base, roughly pH 10 — something like ammonia or a strong cleaner.